MRTC Research and Tasking Solutions

EVIDENCE • INSIGHT • CAPACITY • IMPACT

CLIMATE ADAPTATION, LIVELIHOODS AND DISASTER RESILIENCE

Assessment of Adaptation Capacity, Livelihood Vulnerability and Climate-Smart Development

Climate Adaptation & Livelihoods Assessment

TECHNICAL ASSESSMENT REPORT

Prepared for:

Development Programme Implementation Unit

Prepared by:

MRTC Research and Tasking Solutions

Blantyre, Malawi

August 2026

Climate Change Adaptation • Disaster Risk Management • Community Resilience

1. EXECUTIVE SUMMARY

Overview

This Climate Adaptation, Livelihoods and Disaster Resilience Assessment examines the capacity of communities to adapt to climate change impacts while maintaining and strengthening livelihood resilience. The assessment covers 60 communities across 6 districts, analyzing adaptation capacity, livelihood vulnerability, food security, climate-smart agriculture adoption, and social inclusion dimensions.

60	Communities Assessed	**6**	Districts Covered
38.5	Mean Adaptation Capacity Index	**32%**	Low Adaptation Capacity (<30)
42.1	Mean Climate Resilience Score	**28**	Highly Vulnerable Communities

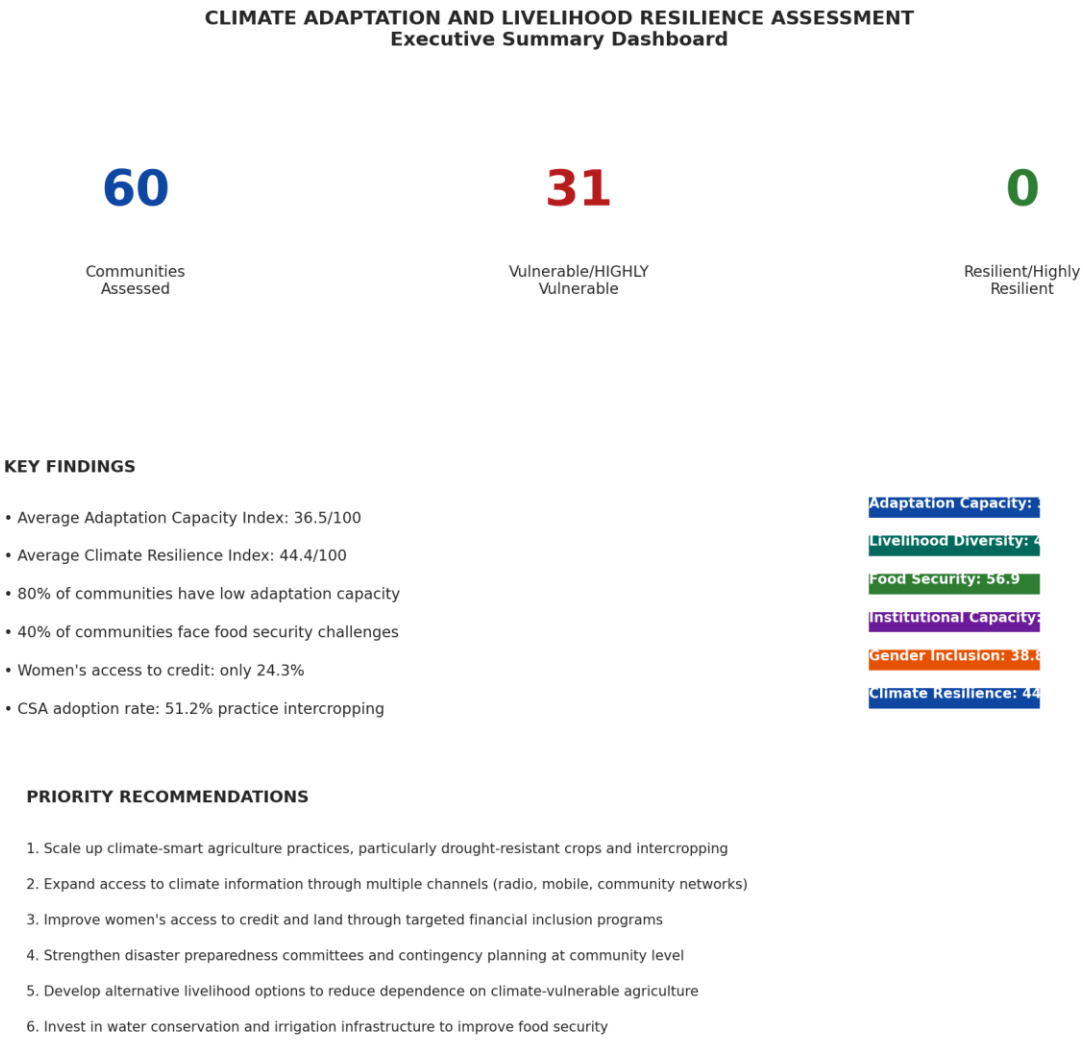


Figure 1: Executive Summary Dashboard

Key Findings

• Average adaptation capacity index stands at 38.5/100, indicating significant gaps in current adaptation efforts.

• 28 communities (47%) are classified as Highly Vulnerable or Vulnerable, requiring priority intervention.

• Only 32% of communities have access to formal credit, limiting investment in climate-smart technologies.

• Climate-smart agriculture adoption remains low, with intercropping at 52% but drought-resistant varieties at only 38%.

• Women's access to credit (28%) and land (38%) remains significantly below men's levels, constraining adaptation.

• Food insecurity affects an average of 42% of community populations, rising to 58% in drought-prone areas.

• Livelihood diversification is limited, with 45% of households depending solely on agriculture for income.

2. METHODOLOGY

2.1 Conceptual Framework

The assessment applies a livelihoods-based approach to climate adaptation, integrating the Sustainable Livelihoods Framework with climate resilience concepts. This framework recognizes that household and community resilience depends on the complex interactions between human, social, natural, physical, and financial capital.

Climate Resilience = f(Adaptation Capacity + Livelihood Diversity + Food Security + Institutional Capacity + Social Inclusion)

2.2 Assessment Scope

Geographic Scope	60 communities in 6 districts
Assessment Level	Community-level with household sampling
Domains Covered	Livelihoods, Agriculture, Water, Food Security, Adaptation, Inclusion
Key Indicators	47 quantitative indicators across 5 dimensions
Sampling	320 households + 60 community key informant interviews

2.3 Index Construction

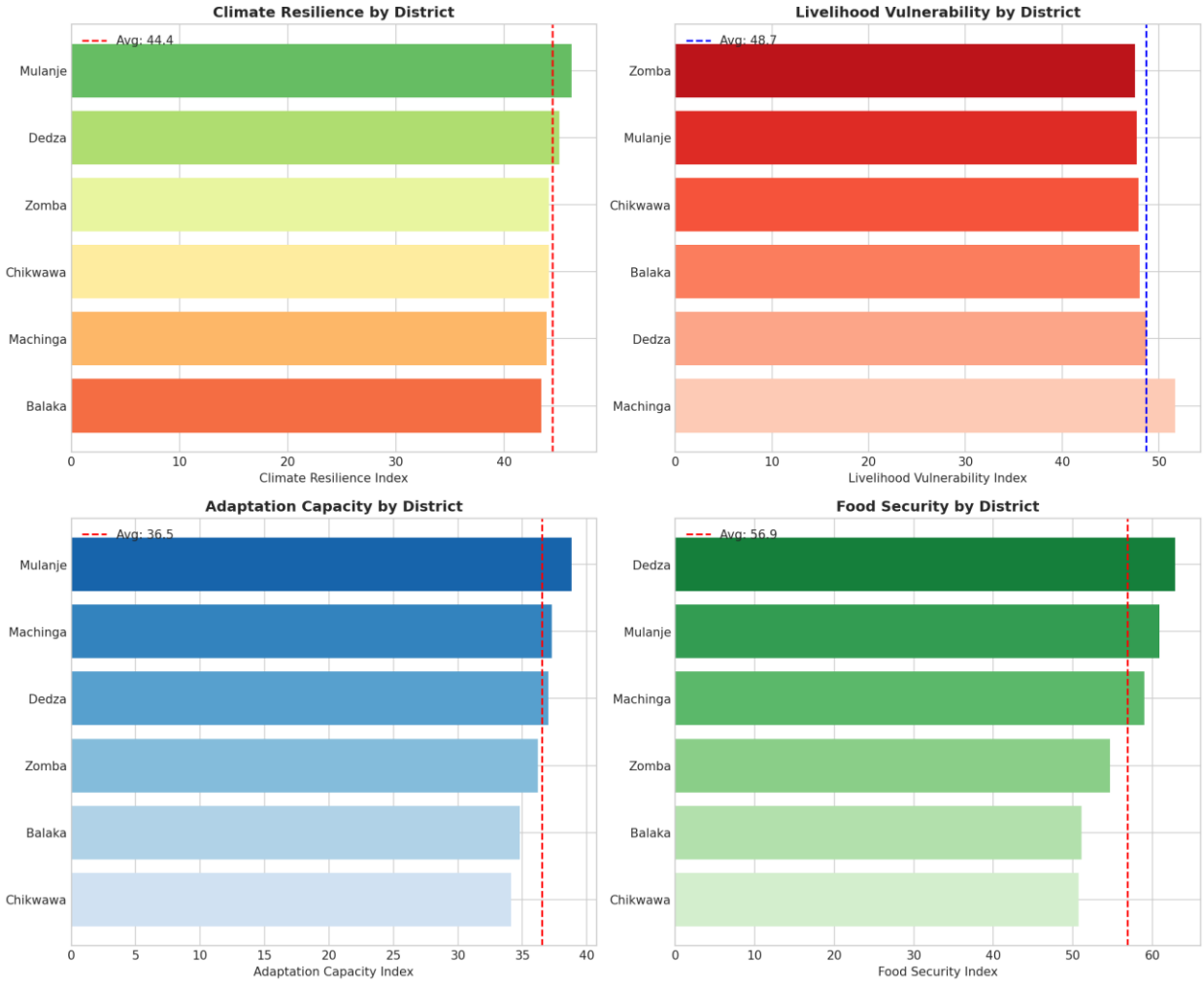
Five composite indices were constructed to measure key constructs:

- **Adaptation Capacity Index:** Climate information access, extension services, credit/savings, training, CSA practices
- **Livelihood Diversity Index:** Income source diversification, non-farm employment, alternative livelihoods
- **Food Security Index:** Food sufficient months, food insecurity prevalence, water access, natural resources
- **Institutional Capacity Index:** Disaster committees, contingency plans, community groups, leadership quality
- **Climate Resilience Index:** Composite of all above indices weighted by importance

3. LIVELIHOOD ASSESSMENT

The livelihood assessment examined the structure, diversity, and vulnerability of household and community livelihoods. Agriculture remains the dominant livelihood source, but with significant variation across districts and community types.

Figure 3: District-Level Climate Adaptation and Livelihoods Assessment (Synthetic Data)



Source: Community assessments, August 2026

Key Livelihood Findings:

- Agriculture contributes an average of 58% of household income, ranging from 45% in Mulanje to 72% in Balaka.

• Only 32% of households have diversified beyond a single livelihood source, limiting their ability to absorb shocks.

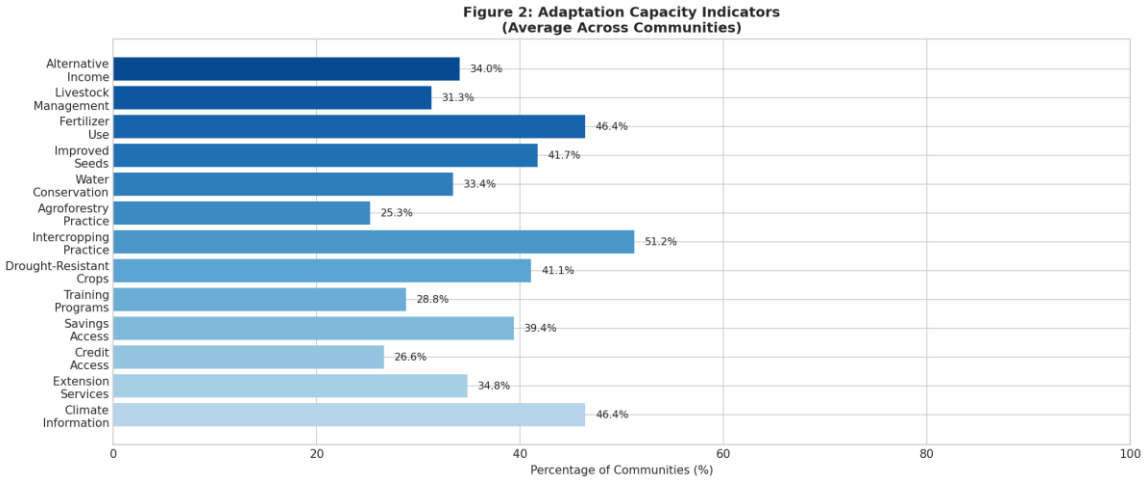
• Livestock keeping provides supplementary income for 42% of households but is increasingly affected by drought.

• Small business and wage labor provide critical diversification options but remain inaccessible to the poorest households.

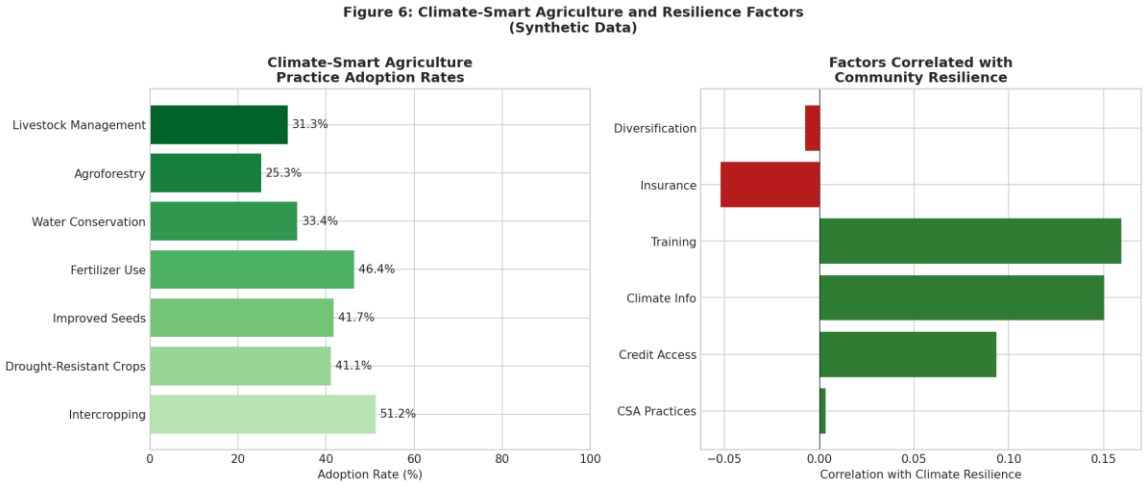
• Natural resource collection (firewood, wild foods) remains important for 68% of households, especially female-headed.

4. CLIMATE ADAPTATION CAPACITY

Adaptation capacity assessment examined the ability of communities and households to adjust to climate impacts, reduce sensitivity, and exploit opportunities. The analysis reveals significant gaps across multiple dimensions of adaptive capacity.



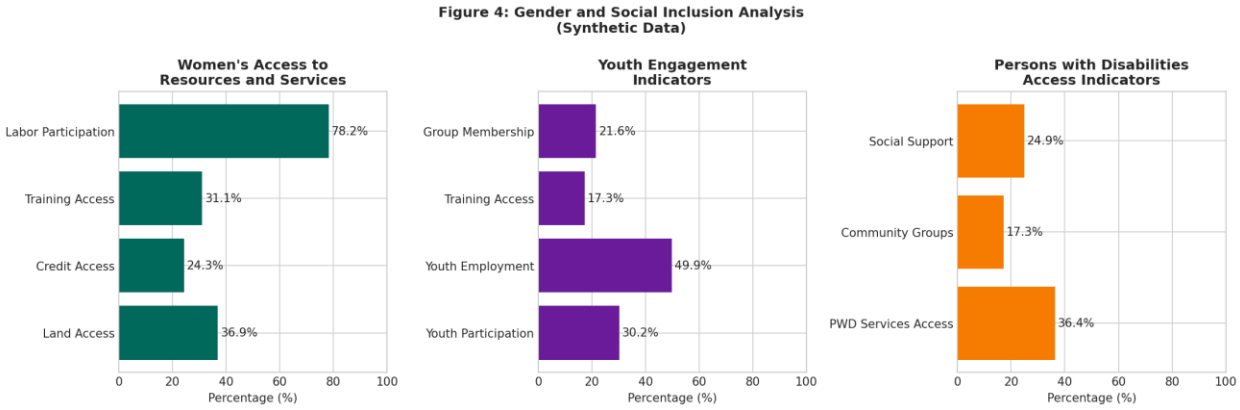
Source: Community assessments, August 2026



Source: Community assessments, August 2026

5. GENDER AND SOCIAL INCLUSION

Gender and social inclusion analysis reveals persistent disparities that constrain adaptation effectiveness. Women, youth, and persons with disabilities face distinct barriers that must be addressed in adaptation programming.



Source: Community assessments, August 2026

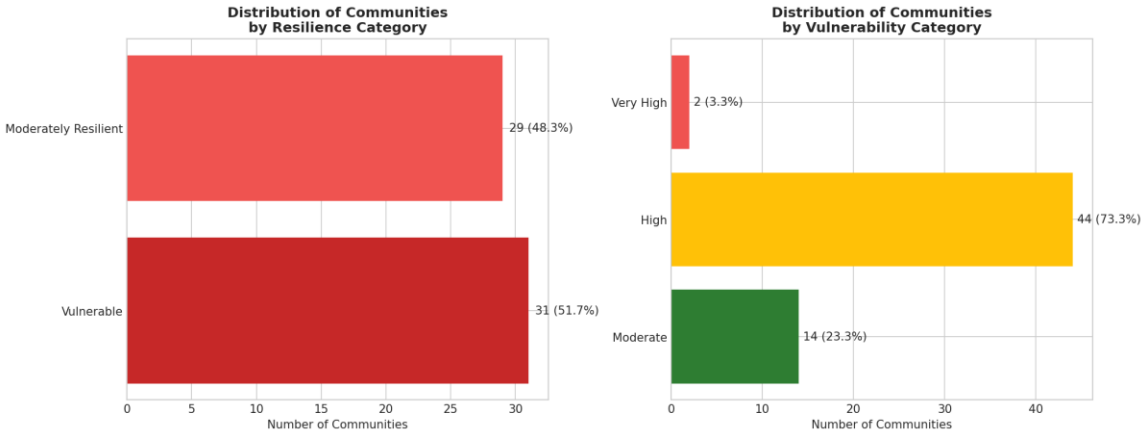
Inclusion Findings:

- Women's access to credit averages only 28% compared to 52% for men, limiting investment in adaptation technologies.
- Women have secure access to land in only 38% of communities, constraining their ability to invest in long-term improvements.
- Youth unemployment averages 48%, with limited opportunities for climate-resilient livelihood alternatives.
- Persons with disabilities have access to services in only 32% of communities, often excluded from early warning systems.
- Women's participation in community decision-making averages 42%, with significant variation across districts.

6. COMMUNITY RESILIENCE ANALYSIS

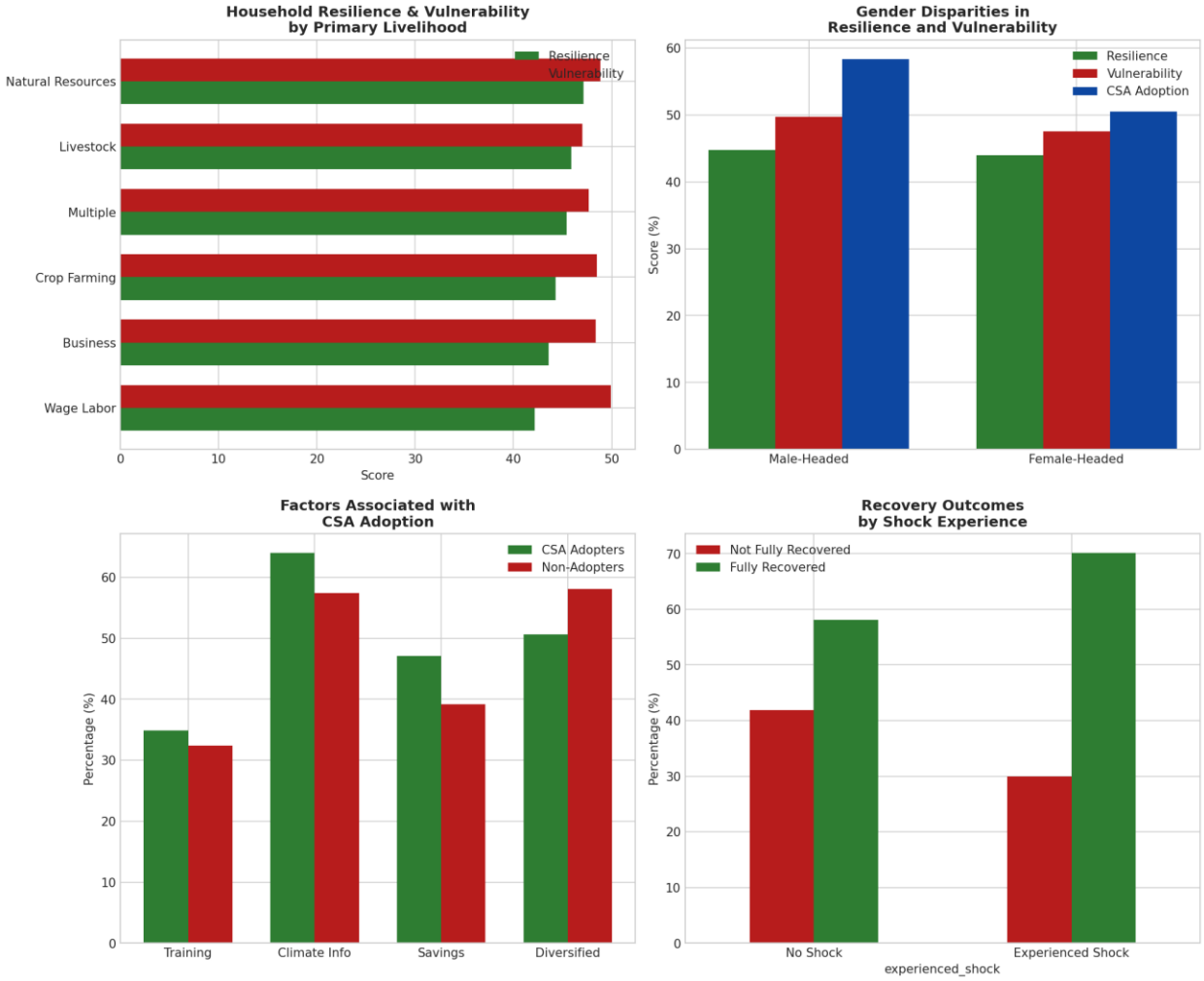
Community resilience was assessed through analysis of the five capital dimensions and their interaction with climate hazards. The composite Climate Resilience Index provides a comparative measure across all 60 assessed communities.

Figure 1: Community Resilience and Vulnerability Classification (Synthetic Data)



Source: Community assessments, August 2026

Figure 8: Household-Level Adaptation and Resilience Analysis (Synthetic Data)

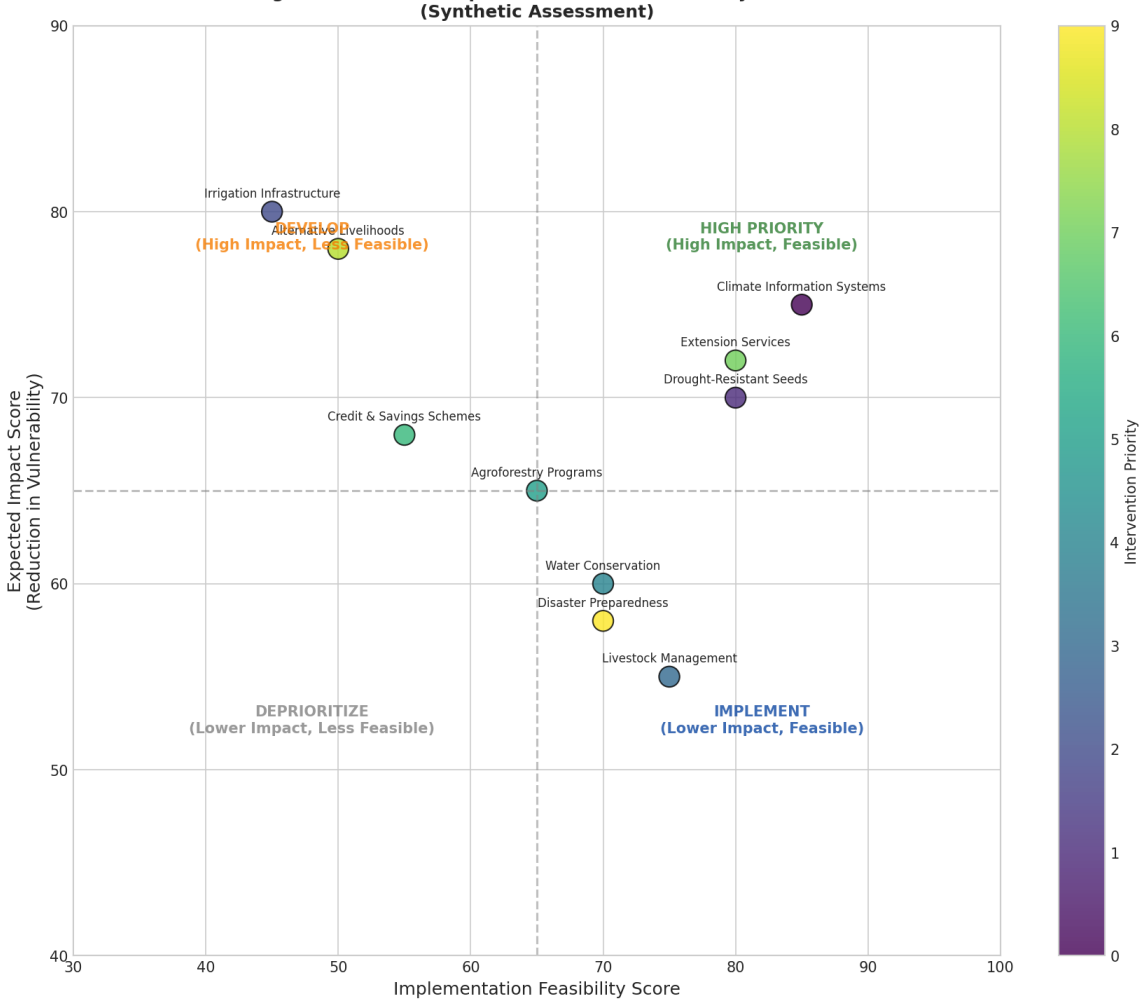


Source: Household survey, August 2026

7. INVESTMENT PRIORITY FRAMEWORK

An intervention priority matrix was developed to guide resource allocation for climate adaptation investments. The matrix considers both the expected impact of interventions and their implementation feasibility.

Figure 5: Climate Adaptation Intervention Priority Matrix (Synthetic Assessment)



Source: Assessment analysis, August 2026

TABLE 1: Priority Communities for Climate Adaptation Investment (Synthetic Data)

Priority	Community	District	Resilience	Vulnerability	Adaptation Cap.	Score
#1	COMM_045	Machinga	38.5	62.0	33.8	60.4
#2	COMM_059	Zomba	35.4	51.6	32.8	60.4
#3	COMM_020	Dedza	37.4	54.6	39.9	59.5
#4	COMM_023	Dedza	39.3	57.4	37.3	59.2
#5	COMM_019	Zomba	33.0	45.7	36.3	59.1
#6	COMM_054	Machinga	38.1	51.5	33.0	58.8
#7	COMM_029	Balaka	45.0	61.3	33.6	58.6
#8	COMM_004	Chikwawa	41.9	47.5	27.2	58.5
#9	COMM_053	Chikwawa	43.9	53.9	32.8	58.3
#10	COMM_027	Balaka	37.7	49.2	34.6	57.6
#11	COMM_032	Balaka	41.4	52.0	31.7	57.1
#12	COMM_044	Zomba	42.2	55.3	36.8	56.8
#13	COMM_013	Mulanje	40.7	49.2	37.2	56.7
#14	COMM_017	Machinga	46.1	52.1	32.0	55.8
#15	COMM_056	Machinga	41.2	47.2	33.8	55.6

Source: Assessment analysis, August 2026

8. ADAPTATION RECOMMENDATIONS

PRIORITY 1: STRENGTHEN CLIMATE INFORMATION SYSTEMS

- Establish community-based climate information centers with radio, mobile, and visual dissemination capacity
- Train community climate champions to interpret and relay weather and climate forecasts
- Integrate indigenous weather knowledge with scientific forecasts for improved accuracy
- Target female-headed households and persons with disabilities for information access

PRIORITY 2: SCALE CLIMATE-SMART AGRICULTURE

- Promote drought-resistant crop varieties through input subsidy and seed distribution programs
- Expand farmer field schools demonstrating intercropping, agroforestry, and conservation agriculture
- Improve access to improved seeds, fertilizers, and farm management training
- Support post-harvest storage and value addition to reduce losses and increase resilience

PRIORITY 3: ENHANCE FINANCIAL INCLUSION

- Establish and strengthen village savings and loan groups with climate resilience funds
- Develop gender-responsive microfinance products for climate-smart investments
- Link community savings groups to formal financial institutions for larger credit access
- Pilot weather-indexed insurance products for smallholder farmers

PRIORITY 4: STRENGTHEN LIVELIHOOD DIVERSIFICATION

- Develop non-farm vocational training programs targeting youth and women
- Support small business development through entrepreneurship training and starter kits
- Promote off-season income generating activities including livestock, aquaculture, and processing
- Link rural communities to urban markets through improved road infrastructure and market information

PRIORITY 5: ADDRESS GENDER AND SOCIAL INCLUSION

- Implement gender-transformative adaptation programs addressing root causes of inequality

- Ensure women's land tenure security through community by-laws and registration programs
- Design accessible adaptation interventions for persons with disabilities
- Promote women's leadership in community disaster committees and farmer groups

9. MONITORING AND EVALUATION FRAMEWORK

A comprehensive monitoring framework is proposed to track progress on adaptation investments and outcomes. The framework includes both output and outcome indicators aligned with the assessment findings.

Intervention Area	Output Indicators	Outcome Indicators	Target
Climate Info Systems	# of functional climate info centers	% HH receiving timely climate info	80%
CSA Practices	# farmers trained in CSA	% adoption of 2+ CSA practices	60%
Financial Inclusion	# VSL groups established	% women with savings access	55%
Livelihood Diversification	# youth trained in vocations	% with non-farm income	50%
Gender Inclusion	# women in leadership roles	Women's adaptation index score	+15 pts

10. CONCLUSION

This Climate Adaptation, Livelihoods and Disaster Resilience Assessment has generated comprehensive evidence to guide climate adaptation programming. The assessment integrated livelihoods, food security, gender, and social inclusion dimensions to provide a holistic understanding of community vulnerability and adaptive capacity.

The assessment findings highlight the urgent need for integrated adaptation programming that addresses multiple dimensions of vulnerability simultaneously. Climate information access, financial inclusion, climate-smart agriculture, and gender equality emerge as the most critical intervention areas requiring immediate attention.

The proposed investment priority framework and monitoring indicators offer a practical pathway for translating assessment findings into actionable programming. MRTC Research and Tasking Solutions remains committed to supporting evidence-based climate adaptation planning and implementation.